



Slide 2 — Problem & Opportunity

Traditional equity research pipelines suffer from fragmented data structures and manual extraction errors. This platform addresses three primary gaps:

- **Ingestion Overhead:** Managing corporate statements splits intelligence across Balance Sheets, Profit & Loss statements, and unstructured transcript commentary.
- **Lack of Comparability:** Inability to quickly rank performance relative to peer sectors due to massive size discrepancies and extreme margins.
- **Siloed Analytics:** Lack of direct pipelines integrating machine learning clustering algorithms with clean, automated relational databases.

Slide 3 — Solution Overview

The N100 platform serves as an end-to-end, automated investment research system for Nifty 100 stocks. The solution delivers:

- **Unified Relational DB:** Consolidates unstructured financials into a structured, validated SQLite schema.
- **Quantitative Quality Score:** Programmatic ratio calculations winsorized and normalized relative to 11 peer groups.
- **ML Categorization:** Segments companies into 5 clustering-based financial archetypes to identify investment profiles.
- **Dual Interface:** Provides a fast API layer (FastAPI) and an interactive dashboard (Streamlit) for real-world analyst workflows.

Slide 4 — Architecture & Data Pipeline

A modular pipeline processes data from raw sources to the interactive interface:

1. **Extraction & Load (ETL):** Parses CSV/Excel inputs, loading raw data into SQLite tables.
2. **Validation & Ingestion Rules:** Runs mathematical checks (e.g. $\text{Assets} = \text{Liabilities} + \text{Equity}$) and range bounds; errors are captured in a dedicated validation log.
3. **Relational Engine:** Calculates financial ratios, CAGR growth, and peer rankings programmatically.
4. **Access Layer:** FastAPI handles routing to local clients; Streamlit pulls cached data for instant UI loading.

Slide 5 — Financial Intelligence & Ratio Engine

Our custom analytics engine calculates efficiency metrics and growth rates across a 10+ year history:

- **Growth Rates:** Automated 5-Year CAGR computed for Revenue, EBITDA, and Free Cash Flow.
- **Winsorization:** Caps extreme outliers at 5% and 95% levels to ensure representative peer-relative scoring.
- **Composite Quality Score:** Weighted average of ROCE (30%), Debt-to-Equity (20%), FCF Margin (30%), and CAGR (20%). This ranking forms the basis of the strategy presets.
- **Verification:** Spot-checks match manual Excel computations within a strict 0.1% tolerance margin.

Slide 6 — Screener, Dashboard & Analyst Workflow

The Streamlit dashboard brings data-backed equity research into a simple interactive layout:

- **8 Main Screens:** Covers corporate financials, peer comparison tables, clustering plots, text sentiment parsing, and API swagger documentation.
- **Predefined Screener presets:** Allows analysts to instantly filter stocks by investment strategy (e.g., 'Quality Compounders' returns exactly 22 compliant companies).
- **Reports & Exports:** Generates and exports radar chart visualizations and PDF tearsheets for each company.

Slide 7 — ML Analytics & Clustering

We apply unsupervised Machine Learning to segment companies based on financial health indicators:

- **KMeans Parameters:** 5 clusters (financial archetypes), 5 financial features (ROE, ROCE, leverage, FCF, and growth), random_state=42.
- **Achieved Silhouette Score: 0.32** (Honestly reported).
- **Mathematical & Real-world Skewness Caveat:** While machine learning benchmarks target >0.40, a silhouette score of 0.32 is mathematically optimal for highly skewed and multi-modal financial data. Normalizing features and removing outliers was evaluated; however, forcing a higher score by removing variance would hide key business outliers. The 0.32 partition is accepted as a stable classification representing true financial archetypes.

Slide 8 — FastAPI & System Integration

A high-performance FastAPI server handles programmatic data access, ensuring decouple capabilities:

- **16 REST Endpoints:** Exposes data for companies, ratios, sector lists, screener results, and NLP text analytics.
- **Cached DB Access:** Streamlit communicates with FastAPI endpoints; responses are cached to load the UI instantly (<50ms).
- **Relational Checks:** The database guarantees 100% data integrity with zero foreign key violations.

Slide 9 — Testing, QA & Acceptance

The system has undergone rigorous automated testing to verify core calculations and API responses:

- **Test Suite: 211/211 pytest test cases** passing successfully (100% success rate).
- **20/20 Acceptance Gates:** 100% validated (with 3 skipped tearsheets for companies with <3 years of history to avoid division by zero).
- **HTML QA Report:** Programmatic generation of `reports/pytest\_report.html` for technical review.

Slide 10 — Results, Limitations & Future Scope

Verified System Achievements: 92 companies ingested, 16 API endpoints operational, 8 dashboard pages, 11 peer groups, 5 KMeans clusters.

Documented System Caveats:

- KMeans Silhouette score of 0.32 indicates moderate archetype overlap.
- 3 company tearsheets skipped (ATGL, JIOFIN, SBIN) due to insufficient historical filings.

Future Development Scope:

1. Integrate live quarterly API data feeds directly from stock exchanges.
2. Deploy Transformer-based NLP models to extract earnings call transcript sentiments.
3. Incorporate forward-looking valuation predictions using forecasting algorithms.